Open-Vocabulary RGB-Thermal Semantic Segmentation

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Abstract. RGB-Thermal (RGB-T) semantic segmentation is an important research branch of multi-modal image segmentation. The current RGB-T semantic segmentation methods generally have two unsolved and typical shortcomings. First, they do not have the open-vocabulary recognition ability, which significantly limits their application scenarios. Second, when fusing RGB and thermal images, they often need to design complex fusion network structures, which usually results in low network training efficiency. We present OpenRSS, the **Open**-vocabulary **R**GB-T **S**emantic **S**egmentation method, to solve these two disadvantages. To our knowledge, OpenRSS is the first RGB-T semantic segmentation method with open-vocabulary segmentation capability. OpenRSS modifies the basic segmentation model SAM for RGB-T semantic segmentation by adding the proposed thermal information prompt module and dynamic low-rank adaptation strategy to SAM. These designs effectively fuse the RGB and thermal information, but with much fewer trainable parameters than other methods. OpenRSS achieves the open-vocabulary capability by jointly utilizing the vision-language model CLIP and the modified SAM. Through extensive experiments, OpenRSS demonstrates its effective open-vocabulary semantic segmentation ability on RGB-T images. It outperforms other state-of-the-art RGB open-vocabulary semantic segmentation methods on multiple RGB-T semantic segmentation benchmarks: +12.1% mIoU on the MFNet dataset, +18.4% mIoU on the MCubeS dataset, and +21.4% mIoU on the Freiburg Thermal dataset. Code will be released at https://github.com/SXDR/OpenRSS.

Keywords: Open-vocabulary · RGB-T semantic segmentation · Prompt learning

1 Introduction

Existing deep learning-based semantic segmentation methods [22, 25, 41] mainly rely on single-modal RGB images, which are effective only under well-lit conditions during the daytime and perform poorly in challenging lighting conditions

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(e.g., nighttime, foggy weather). With the popularity of thermal cameras, researchers have recently started incorporating thermal images as a supplementary modality into RGB semantic segmentation [8], known as RGB-T semantic segmentation.

With the advancement of deep learning, current RGB-T semantic segmentation models mainly rely on supervised training with a predefined set of fixed categories, which cannot generalize to categories beyond the training set. This shortcoming significantly limits the application scenarios of current RGB-T semantic segmentation methods. In contrast, humans understand scenes in an open-vocabulary manner and recognize around 30,000 basic categories [1, 7]. Moreover, humans can further identify new categories through language descriptions.

Recently, large-scale vision-language models, represented by CLIP [26], achieve image-level classification across arbitrary categories, i.e., open-vocabulary classification. However, since the CLIP model is trained using image-level contrastive learning, how to effectively utilize CLIP in semantic segmentation is still an open problem. Some researchers adopt a two-stage training paradigm [5, 20, 37]: first, using a model to generate category-agnostic mask proposals, and then a pre-trained CLIP model for open-vocabulary classification on each mask. While others choose to train a side network [36] to integrate pre-trained CLIP model features into the side network for mask recognition. The promising performance of CLIP prompts us to explore its applicability in RGB-T semantic segmentation. But unlike the methods mentioned above, in this paper, we jointly utilize the basic segmentation model SAM [16] and CLIP through an elegant way to achieve open-vocabulary semantic segmentation, which effectively takes advantage of well-learned visual features and high segmentation performance of SAM and powerful open-vocabulary classification ability of CLIP.

While SAM has learned a powerful and general visual segmentation model using the largest segmentation dataset to date, SA-1B, it is still unclear how to effectively transfer its knowledge from RGB images to RGB-T images. The most direct strategy is to train the SAM from scratch, specifically for RGB-T images. However, on the one hand, full model fine-tuning is time-consuming, inefficient, and burdensome regarding parameter storage. On the other hand, with the limited quantity of RGB-T semantic segmentation datasets, it is easy to produce over-fitting results. Thus, a natural question arises: is there a more efficient and effective method to enable SAM models trained on RGB images to be utilized for RGB-T images? Our answer to this question is affirmative, and we provide a solution with the paradigm of prompt learning. Prompt learning, also known as prompt tuning, originated from natural language processing (NLP) [17, 19], aims to inject text prompts into language models to transfer the knowledge of foundation models to various downstream tasks. Inspired by this, researchers apply prompt tuning to visual tasks [3, 6, 13, 33, 42, 43, 45] by freezing the entire foundation model and adding some learnable parameters only to the input side to learn effective visual prompts. Intuitively, there is a significant correlation between multimodal and RGB modalities, and they should share most of the prior knowledge about feature extraction. Following this, we propose OpenRSS, the first open-vocabulary RGB-T semantic segmentation method.

Unlike other RGB-T semantic segmentation methods that always need complex fusion networks to fuse RGB and thermal information, which inevitably adds considerable computation and training burden to their approaches. In OpenRSS, only a small amount of parameters are trainable, which makes the training of OpenRSS quite efficient and also decreases the risk of over-fitting. During the training, OpenRSS freezes the entire image encoder of SAM and only learns visual prompts based on RGB and thermal modality. Specifically, OpenRSS freezes the image encoder of SAM and adds several simple and efficient Thermal Information Prompt (TIP) modules to SAM to produce visual prompts to learn the complementarity between RGB and thermal modality effectively. Then, in the mask decoder of SAM, we align pixel embeddings with the text embeddings corresponding to the semantic classes generated by CLIP. Due to the flexible label representations provided by the text encoder, OpenRSS can perform open-vocabulary segmentation without requiring additional training samples. Additionally, we employ a Dynamic Low-Rank Adaptation (DLoRA) strategy to fine-tune the image encoder of SAM. For fairness and reproducibility, our research is based on the officially released SAM and CLIP models. The main contributions of this paper are as follows:

- We propose OpenRSS, the first open-vocabulary RGB-T semantic segmentation model, where, with the aid of prompt learning and dynamic low-rank adaptation strategy, existing foundation models can effectively adapt to open-vocabulary RGB-T semantic segmentation.
- We introduce a Thermal Information Prompt (TIP) module that generates simple yet efficient visual prompts to fuse RGB and thermal information, avoiding the complex and redundant additional branches designed in previous methods.
- We employ a Dynamic Low-Rank Adaptation (DLoRA) strategy to optimize a series of parameters in the image encoder of SAM.
- Extensive experiments demonstrate that OpenRSS outperforms other state-of-the-art RGB open-vocabulary semantic segmentation methods on multiple RGB-T semantic segmentation benchmarks: +12.1% mIoU on the MFNet dataset, +18.4% mIoU on the MCubeS dataset, and +21.4% mIoU on the Freiburg Thermal dataset, while maintaining parameter efficiency (only 1% trainable parameters).

The remainder of this article is organized as follows. We review related work in Sec. 2. In Sec. 3, we provide a detailed description of our network architecture. Sec. 4 present experimental results and discussions. Finally, we summarize the work in Sec. 5.

2 Related Work

2.1 RGB-T Semantic Segmentation

For complex scenes with challenging lighting conditions, thermal images are highly complementary to RGB images. Existing RGB-T semantic segmentation methods focus on designing complex cross-modal fusion paradigms. We briefly review RGB-T semantic segmentation from three paradigms: encoder fusion, decoder fusion, and feature fusion. For the encoder fusion paradigm, RTFNet [28] and FusegSeg [29] integrate thermal features into RGB features by element-wise summation. FEANet [4] introduces a two-stage feature-enhanced attention network, which utilizes channel and spatial attention to fuse RGB and thermal information in a complementary way. For the decoder fusion paradigm, MFNet [8] uses two identical encoders to extract RGB and thermal features and feed them into the decoder for integration. For the feature fusion paradigm, EGFNet [44] embeds prior edge information into feature maps for cross-modal fusion. ABMDRNet [40] employs a bi-directional image-to-image translation based method to reduce the modality differences, followed by channel-wise weighted fusion for final prediction. CMX [38] first combines features from other modalities in both spatial and channel directions to calibrate the current modality's features, then fuses them for cross-modal fusion.

Unlike the methods mentioned above, which always have to train complex additional network branches to extract thermal-related features, OpenRSS only needs to learn a small amount of trainable parameters to inject thermal-related information into the image encoder of SAM, whose parameters are frozen during training. This merit gives OpenRSS high parameters and training efficiency.

2.2 Large-scale Models

The Transformer's robust scalability allows large-scale models to have hundreds of billions of parameters. Large-scale models, such as BERT [14], InstructGPT [24], GPT-4 [23], etc, achieve groundbreaking progress in natural language processing tasks. Large-scale models like SAM [16] and SegGPT [34] demonstrate outstanding zero-shot generalization capabilities in computer vision. Recently, large-scale vision-language models represented by CLIP [26] and ALIGN [12] can recognize arbitrary categories described in the text, known as open-vocabulary classification.

Our work further explores how to effectively utilize pre-trained visual and visual-language large-scale models to perform open-vocabulary semantic segmentation on RGB-T images, and accordingly proposes our solution: the OpenRSS model.

2.3 Visual Prompt Learning

In computer vision, downstream tasks are always executed by fine-tuning large pre-trained models comprehensively for a long time. Specifically, different downstream data are used to retrain downstream tasks to update all parameters. Although this approach yields satisfactory performance, it is inefficient regarding parameter usage and ultimately loses the universal knowledge learned by the pre-trained models. Prompt learning (also known as prompt tuning) is recognized as a trade-off between the training burden and the final performance in the downstream tasks and has recently garnered widespread attention. It initially shone in natural language processing before being introduced to various computer vision tasks. For example, VPT [13] presents a set of learnable parameters for Transformer encoders and significantly outperforms full fine-tuning on multiple downstream tasks. ViPT [45] introduces prompts for pre-trained single-modal trackers, making them applicable to multimodal tracking tasks. UniSOD [33] designs a switchable prompt generation block to interact with single-modal or multimodal inputs.

In previous works, they often used simple 1×1 or 3×3 convolutions to generate prompts. In contrast, we specifically design a Thermal Information Prompt (TIP) module with a Transformer block inside to generate prompts for dense prediction, enabling the pre-trained SAM model to adapt to RGB-T images.

2.4 Open-vocabulary Semantic Segmentation

Open-vocabulary semantic segmentation refers to identifying pixel-level categories in images through arbitrary text descriptions. In initial works like ZS3Net [2], generative models are used to synthesize pixel-level features through word embeddings of unseen classes. GroupVit [35] introduces a grouping mechanism to cluster images automatically under text supervision. With the remarkable performance of pre-trained large-scale vision-language models in open-vocabulary classification, researchers begin exploring their performance in open-vocabulary semantic segmentation. These approaches can be broadly categorized into the following three types: 1)LSeg [18] aligns pixel-level embeddings with corresponding semantic class text embeddings for pixel-level classification. 2) [5, 20, 37] adopt a two-stage segmentation framework, where MaskFormer first generates a set of category-agnostic mask proposals, followed by CLIP performing open-vocabulary classification on each mask. 3)SAN [36] designs a side network to enable CLIP to perceive the predictions and recognition of masks.

Unlike these methods, OpenRSS is an end-to-end network that elegantly utilizes SAM and CLIP to achieve open-vocabulary semantic segmentation.

3 Method

In this paper, we present OpenRSS, an end-to-end framework that combines pre-trained large-scale models SAM and CLIP to perform open-vocabulary RGB-T semantic segmentation. Instead of fully fine-tuning all the parameters of the models, OpenRSS only learns a small amount of prompt information. Furthermore, we introduce trainable dynamic LoRA bypasses to adjust the frozen image encoder.

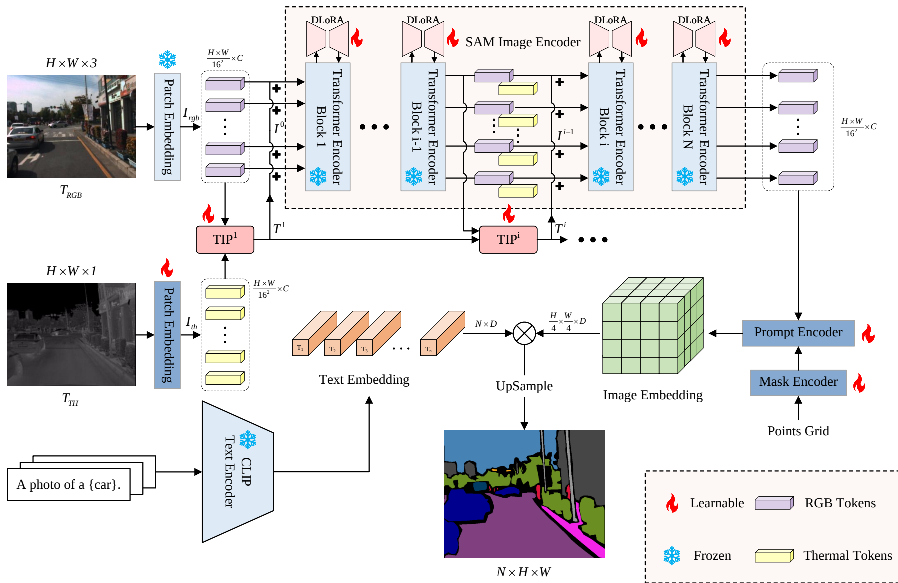

Fig. 1: Overview architecture of our OpenRSS. It inputs a pair of RGB and thermal images and generates corresponding RGB and thermal tokens through patch embeddings. Then, it utilizes the frozen Transformer encoder blocks of SAM for feature extraction. The TIP modules are inserted into the image encoder of SAM to learn effective visual prompts. DLoRA further adjusts the frozen image encoder. Finally, the mask decoder aligns image and text embeddings to generate the final segmentation map.

3.1 Overview

The overall architecture of our OpenRSS is shown in Fig. 1. Formally, OpenRSS incorporates an additional thermal image input stream instead of a thermal feature extraction branch, which is fed into the network simultaneously with the RGB stream. Given a pair of RGB and thermal images, OpenRSS first projects the two information streams (T_{RGB} and T_{TH}) into patch embedding layers separately. After projection, we obtain two C-dimensional embeddings, which we denote as RGB tokens $\boldsymbol{I}_{rgb}$ and thermal tokens $\boldsymbol{I}_{th}$, respectively. Then, $\boldsymbol{I}_{rgb}$ and $\boldsymbol{I}_{th}$ are input into the TIP module to generate thermal information prompts. The learned prompt information is added to the original RGB information stream $\boldsymbol{I}_{rgb}$ in a residual manner before being fed into the fully frozen image encoder of SAM:

$$\boldsymbol{I}^i = \boldsymbol{I}^i_{rgb} + \boldsymbol{T}^{i+1}, \quad i = 0, 1, \ldots, L-1, \tag{1}$$

where $\boldsymbol{I}^i$ is the final sequence of image tokens fed into the Transformer layers of the encoder, and $\boldsymbol{T}^{i+1}$ is the prompt tokens sequence from the $(i+1)$th TIP module. By directly adding the thermal information prompts to the intermediate features generated by SAM, our OpenRSS can effortlessly perform downstream tasks quickly. Furthermore, we also design a trainable dynamic LoRA bypass for each Transformer layer. For each bypass, the features generated by the Transformer layers are first mapped to a low-rank space and then projected back to the initial dimension, aligning it with the original output dimension of the frozen Transformer layers. We use a points grid prompt for the prompt encoder, significantly enhancing SAM's attention to every detail in the image compared to the default embeddings. For the text encoder, we use the frozen text encoder of CLIP. During training, we employ manually crafted prompts for the text encoder, consisting of a natural sentence and category names. However, any sentence can be used during inference to achieve the goal of open-vocabulary. We modify the original mask decoder of SAM to align its dimensionality with the feature embeddings of the text encoder. It simultaneously takes the embeddings from the prompt encoder and the text encoder as input, generating the final segmentation map.

3.2 Thermal Information Prompt

In recent years, following the success of text prompts in NLP, some research has begun to explore the introduction of visual prompts into frozen pre-trained models. Prompt tuning is achieved through some learnable parameters and yields promising results on many downstream visual tasks. A key to our OpenRSS is to generate appropriate thermal information prompts to guide the network in effective open-vocabulary RGB-T semantic segmentation. As shown in Fig. 1, the TIP module is inserted before each Transformer layer. Formally, the thermal information prompts are generated from the token sequences $\{\boldsymbol{I}^0_{rgb}, \boldsymbol{I}^0_{th}\}$ of both modalities. The TIP module learns the complementarity between these two modalities, which can be represented as follows:

$$\boldsymbol{T}^i = TIP^i\left(\boldsymbol{I}^{i-1}, \boldsymbol{T}^{i-1}\right), \quad i = 1, 2, \ldots, L, \tag{2}$$

where TIP^i denotes the ith TIP module. Specifically, $\boldsymbol{I}^0 = \boldsymbol{I}^0_{rgb}$ and $\boldsymbol{T}^0 = \boldsymbol{I}^0_{th}$.

The detailed design of the TIP is shown in Fig. 2. TIP has two input branches, each receiving token sequences from two input streams. Specifically, considering the redundancy of single-modal features and the lightness of the prompt block, we use two 1×1 convolution operations to project each input stream to the size of $[\alpha\times H\times W]$ by:

$$\boldsymbol{Z}_{RGB} = c_1\left(\boldsymbol{I}^{i-1}\right), \quad \boldsymbol{Z}_{TH} = c_2\left(\boldsymbol{T}^{i-1}\right), \tag{3}$$

where α is set to 8 in all TIP modules. $c_1(\cdot)$ and $c_2(\cdot)$ are 1×1 convolution operations. Then, inspired by [32], we use a Transformer block with correlative self-attention for dense prediction to generate enhanced embeddings for $\boldsymbol{Z}^e_{RGB}$, as shown in Fig. 2. The formula for $Correlative\_attn$ is expressed as:

$$Correlative\_attn = \text{Softmax}\left(\boldsymbol{X}\boldsymbol{W}_q\boldsymbol{W}_q^T\boldsymbol{X}^T/\tau\right), \tag{4}$$

where $\boldsymbol{X}$ denotes the input and $\boldsymbol{W}_q$ is the projection matrix. The temperature coefficient τ is by default set to $\sqrt{d}$ following traditional self-attention. Finally,

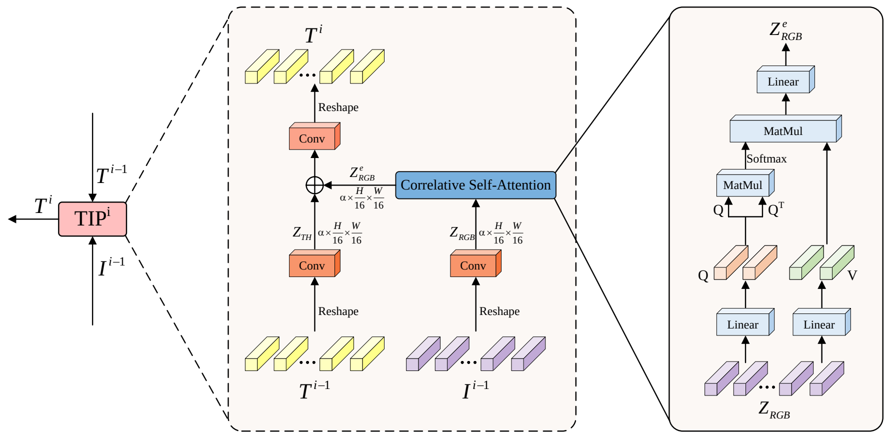

Fig. 2: Overview architecture of our TIP module. Multi-modal input flows are sent separately to the 1×1 convolutions for distilling to the lower-dimension latent space. After passing through the Correlative Self-Attention block, the RGB embeddings are fused with thermal embeddings through element-wise addition. Finally, the multi-modal hybrid embeddings are mapped to the original dimension through a 1×1 convolutional layer.

we obtain the mixed-modal embeddings by element-wise addition and project them back to the original dimension by:

$$\boldsymbol{T}^i = c_3 \left(\boldsymbol{Z}^e_{RGB} + \boldsymbol{Z}_{TH} \right), \tag{5}$$

where $c_3(\cdot)$ is the same as $c_1(\cdot)$ and $c_2(\cdot)$.

3.3 DLoRA in Image Encoder

In OpenRSS, the thermal information prompts generated by TIP are fed into the frozen pre-trained image encoder of SAM to extract features. However, this may not be optimal due to the differences between the RGB and thermal modalities. Inspired by the idea from [10, 30], we design a Dynamic Low-Rank Adaptation (DLoRA) strategy to adjust the frozen image encoder of SAM to address this issue.

Given an input token sequence $\boldsymbol{X} \in \mathbb{R}^{B\times N\times C_{in}}$, the output token sequence $\boldsymbol{X'} \in \mathbb{R}^{B\times N\times C_{out}}$ is obtained through a projection layer $\boldsymbol{W} \in \mathbb{R}^{C_{in}\times C_{out}}$. LoRA assumes the updates to the weights have a low "intrinsic rank" during adaptation, thus proposing a low-rank decomposition to constrain its updates. Following this strategy, the Transformer layer is first frozen to fix $\boldsymbol{W}$, and then a bypass is added to achieve low-rank decomposition. This bypass consists of two linear layers $\boldsymbol{A} \in \mathbb{R}^{C_{in}\times r}$ and $\boldsymbol{B} \in \mathbb{R}^{r\times C_{out}}$, where r $\ll$ min(C_{in}, C_{out}). Therefore, the update of $\boldsymbol{W}$ can be described as:

$$\hat{\boldsymbol{W}} = \boldsymbol{W} + \boldsymbol{\Delta W} = \boldsymbol{W} + \boldsymbol{BA}. \tag{6}$$

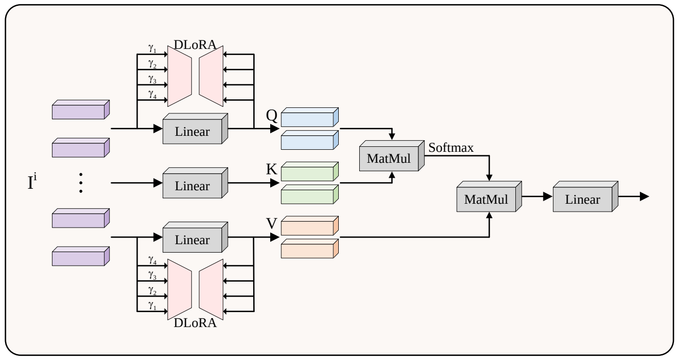

Fig. 3: Overview architecture of our DLoRA. We apply the DLoRA layer to the q and v projection layers of each of the Transformer layers in the image encoder of SAM.

We maintain consistency with [39] and apply LoRA to the query and value projection layers of the multi-head self-attention. Therefore, the modified query and value projection layers will become:

$$\begin{aligned} \boldsymbol{Q} &= \hat{\boldsymbol{W}}_q\boldsymbol{X} = \boldsymbol{W}_q\boldsymbol{X} + \boldsymbol{B}_q\boldsymbol{A}_q\boldsymbol{X}, \\ \boldsymbol{V} &= \hat{\boldsymbol{W}}_v\boldsymbol{X} = \boldsymbol{W}_v\boldsymbol{X} + \boldsymbol{B}_v\boldsymbol{A}_v\boldsymbol{X}, \end{aligned} \tag{7}$$

where $\boldsymbol{W}_q$ and $\boldsymbol{W}_v$ are frozen from image encoder of SAM, and $\boldsymbol{A}_q$, $\boldsymbol{B}_q$, $\boldsymbol{A}_v$ and $\boldsymbol{B}_v$ are trainable parameters.

While LoRA achieves remarkable results, it heavily relies on the choice of rank r, requiring a manual and iterative search for the optimal rank. Therefore, we propose a Dynamic LoRA (DLoRA). Building upon the original LoRA, we increase the number of bypasses (as shown in Fig. 3) from 1 to 4, each with different ranks γ and weights β. Through experimental observation, we find that our DLoRA can surpass the optimal rank found through manual iterative training in just one training.

3.4 Generate Segmentation Map

Prompt Encoder and Mask Decoder. SAM passes the image embeddings generated by its image encoder and the prompts generated by its prompt encoder to its mask decoder for dense prediction. The types of prompts include point, box, and mask. SAM represents the first two sparse prompts as positional encodings and feeds them into the Transformer layers of the mask decoder. SAM performs convolution operations for dense prompts such as masks and then adds them element-wise with the image embeddings.

Due to SAM's prompt mechanism, it is susceptible to the position and quantity of prompt points. Therefore, in OpenRSS, we use a points grid as prompts to ensure that SAM does not overlook any detail in the image. SAM's mask decoder comprises a lightweight Transformer layer and a segmentation head. OpenRSS modifies this segmentation head to align the dimension of the feature map with the dimension of CLIP's text features, enabling pixel-level classification.

Pixel-level Classification via CLIP. Existing CLIP-based open-vocabulary segmentation methods mainly adopt a two-stage region recognition approach. In the first stage, an independent model is trained to generate a set of category-agnostic mask proposals. In the second stage, visual language pre-trained models such as CLIP are used to identify the category of each mask individually. However, this two-stage approach heavily relies on the quality of the masks generated in the first stage. Moreover, the cropped masks, due to missing backgrounds and varying sizes, are not suitable for recognition by CLIP.

To address the problem of region recognition, our OpenRSS adopts pixel-level recognition. We define the feature map outputted by the mask decoder as $\boldsymbol{F} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times D}$, and the class embeddings generated by the CLIP text encoder as $\boldsymbol{E} \in \mathbb{R}^{N \times D}$. Since they have the same embedding dimension, we can directly link them together through the inner product to obtain the final segmentation map:

$$\boldsymbol{S} = \boldsymbol{F} \times \boldsymbol{E}^T, \tag{8}$$

where $\boldsymbol{S} \in \mathbb{R}^{\frac{H}{4} \times \frac{W}{4} \times N}$. This is the standard output of semantic segmentation, hence compatible with mainstream semantic segmentation evaluations.

4 Experiments

4.1 Dataset and Evaluation Metrics

We conduct experiments on 4 datasets: MF dataset [9], KP dataset [11], MCubeS dataset [21] and Freiburg Thermal dataset [31]. All models are trained on the training set of the KP dataset and evaluated on other datasets.

MF Dataset. The dataset consists of urban street scenes captured using an InfRec R500 camera. It includes 1569 pairs of RGB and thermal images, each with a resolution of 480×640 pixels. Among these are 820 pairs of daytime images and 749 pairs of nighttime images. The dataset consists of eight manually labeled classes, as well as a background class.

KP Dataset. The dataset is an RGB-T paired urban driving scene dataset, providing 95328 pairs of RGB and thermal images with a resolution of 512×640. Initially, the KP dataset only provides bounding box labels for object detection tasks. However, fortunately, [15] provides semantic segmentation task labels for 503 daytime and 447 nighttime images. Furthermore, [27] partitions these 950 images into training, validation, and test sets.

MCubeS Dataset. MCubeS is a dataset with pairs of RGB, Near-Infrared (NIR), Degree of Linear Polarization (DoLP), and Angle of Linear Polarization (AoLP) to study semantic material segmentation of 20 classes with a resolution of 1024×1224.

Freiburg Thermal Dataset. The dataset includes 12051 daytime and 8596 nighttime time-synchronized RGB-T image pairs captured in rural and urban environments with a resolution of 650×1920. However, only the test set includes 32 daytime and 32 nighttime images, comprising 12 object classes and one background class.

Evaluation Metrics. We use mean Accuracy (mAcc) and mean Intersection over Union (mIoU) as the evaluation metrics for our model. The formulas for calculation are as follows:

$$mAcc = \frac{1}{n_{cls}} \cdot \sum_i \frac{n_{ii}}{t_i}, \tag{9}$$

$$mIoU = \frac{1}{n_{cls}} \cdot \sum_i \frac{n_{ii}}{t_i + \sum_j n_{ji} - n_{ii}}, \tag{10}$$

where n_{ii} is the number of pixels where class i is predicted as class j, n_{cls} is the total number of target classes (including the background), and $t_i = \sum_j n_{ij}$ is the total number of pixels for target class i (ground truth labels).

4.2 Implementation Details

We conduct all experiments on NVIDIA A100 GPUs. If not specified, we default to using SAM with the ViT-H backbone and CLIP with the ViT-L/16 backbone. We apply DLoRA fine-tuning to the frozen encoder Transformer layers, using four bypasses with ranks selected as 1, 2, 4, and 8, and weights as 0.1, 0.2, 0.4, and 0.3, respectively. All models are trained on the KP dataset with image flipping as a data augmentation technique. We use the AdamW optimizer with a weight decay of 0.01 and an initial learning rate of 5e-4. For the scheduler, we use the WarmupPolyLR scheduler. The loss function consists of DiceLoss and SoftCrossEntropy with weighting coefficients of 0.5 each:

$$DiceLoss = 1 - \frac{2 \sum_{i=1}^{N} y_i \widehat{y_i}}{\sum_{i=1}^{N} y_i^2 + \sum_{i=1}^{N} \widehat{y_i^2}}, \tag{11}$$

where y_i and $\widehat{y_i}$ represent the true and predicted label values of pixel I, respectively, and N represents the total number of pixels.

$$SoftCrossEntropyloss = -\frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{c} \hat{y}_{ij} \log \left(y_{ij}^{d} \right), \tag{12}$$

Table 1: Performance comparison with state-of-the-art methods. For a fair comparison, we reproduce their method with their officially released code. The bold font highlights the best result in each column.

Method	Backbone	MF		MCubeS		FT	
		mAcc↑	mIoU↑	mAcc↑	mIoU↑	mAcc↑	mIoU↑
ZegFormer [5]	R-50	11.11	10.31	10.00	3.43	7.69	0.11
OVSeg [20]	R-101	10.86	9.42	10.09	2.73	8.32	0.98
SAN [36]	ViT-B	14.64	4.13	17.24	10.10	7.13	1.31
ZegFormer [5]	R-101	11.55	10.71	10.46	4.11	7.85	0.79
OVSeg [20]	Swin-B	10.20	6.78	16.49	6.58	10.20	3.74
SAN [36]	ViT-L	3.81	0.38	6.90	4.27	10.60	6.10
OpenRSS(ours)	SAM-B	22.21	18.96	31.76	20.27	16.01	8.14
	SAM-L	23.91	20.08	33.95	24.91	25.33	16.65
	SAM-H	**30.73**	**22.81**	**37.88**	**28.58**	**37.25**	**27.55**

where the n is the number of the batch, $\hat{y}_{ij}$ is a binary indicator if class label c is the correct classification for pixel i, and y_{ij}^{d} is the corresponding predicted probability be normalized to a probability distribution. These two losses are summed to obtain the overall loss function:

$$L_{total} = 0.5 L_{\text{DiceLoss}} + 0.5 L_{\text{SoftCrossEntropyloss}}. \tag{13}$$

4.3 Results and Analysis

Tab. 1 presents the quantitative comparison results of our method with other state-of-the-art open-vocabulary semantic segmentation methods. We only use RGB images as input, rather than pairs of RGB and thermal images, allowing the general open vocabulary semantic segmentation in Tab. 1 to perform RGB-T semantic segmentation. From the table, it can be observed that our method achieves superior performance on both metrics across all three datasets, indicating the effectiveness of our approach. Specifically, our method surpasses other methods with an average of +9.17% mAcc and +6.82% mIoU for SAM-B, +13.57% mAcc and +11.58% mIoU for SAM-L, and +20.46% mAcc and + 17.34% mIoU for SAM-H. This further validates that open-vocabulary semantic segmentation methods trained on RGB images are unsuitable for RGB-T images. Furthermore, we present the qualitative experimental results of OpenRSS on the Freiburg Thermal dataset in Fig. 4.

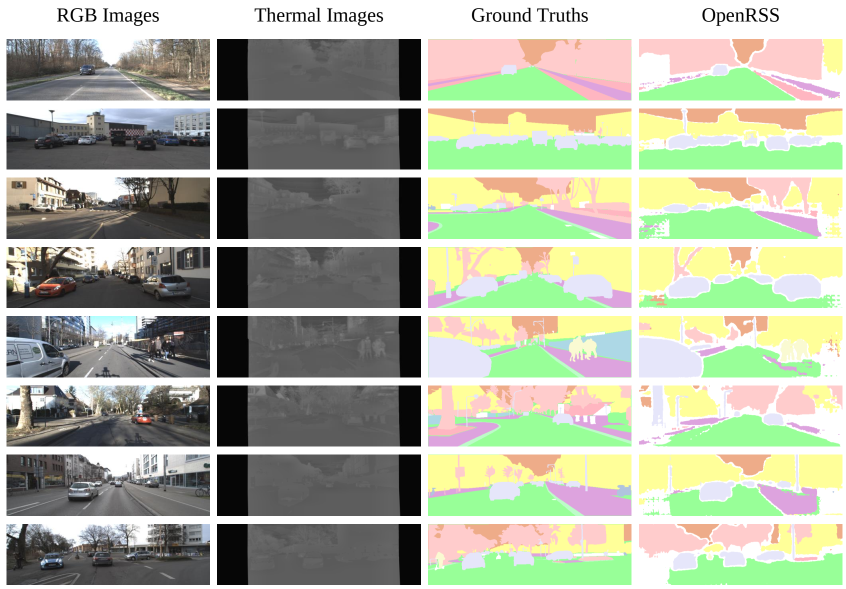

Fig. 4: Qualitative results of OpenRSS on Freiburg Thermal dataset. From left to right: RGB images, Thermal images, ground-truths, OpenRSS results.

Table 2: Ablation studies of TIP module in OpenRSS.

Method	MF	MS	FT
	mIoU↑		
TIP-empty	10.20	16.21	3.90
TIP-shallow	19.98	26.83	25.44
TIP-deep(ours)	22.81	28.58	27.55

Table 3: Ablation studies of attn type in TIP module.

Method	MF	MS	FT
	mIoU↑		
attn-fovea	18.45	23.01	18.58
attn-msa	22.54	21.72	21.56
attn-cor(ours)	22.81	28.58	27.55

4.4 Ablation Study

Importance of TIP. Tab. 2 shows the ablation experiments of the proposed TIP module. To validate the effectiveness of the TIP module, we replace it with additive fusion, resulting in an average mIoU performance drop of 16.21%. Furthermore, we discuss the specific placement of the TIP module. We only position the TIP module before the first Transformer layer, called TIP-shallow. The results indicate that placing the TIP module at each Transformer layer yields the best performance. Finally, we conduct ablation experiments on the correlative self-attention within the TIP module. As shown in Tab. 3, the results demonstrate the effectiveness of correlative self-attention for dense tasks by comparing it to spatial fovea attention and the original multi-head self-attention.

Importance of DLoRA. Tab. 4 shows the ablation experiments of the proposed DLoRA. By manually setting the ranks to 1, 2, 4, and 8, the performance degradation of average mIoU is observed to be 4.69%, 2.55%, 3.03%, and 1.36%, respectively. This demonstrates the significance of DLoRA while also avoiding spending considerable time selecting an optimal low-rank space. More detailed results and analysis, including failure cases, are discussed in the supplementary material.

Table 4: Ablation of rank in DLoRA. *Manually set rank vs. dynamically adaptive rank.* The bold font highlights the best result in each column.

Methods	MF		MCubeS		FT	
	mAcc↑	mIoU↑	mAcc↑	mIoU↑	mAcc↑	mIoU↑
OpenRSS-rank-1	29.20	20.61	37.43	24.69	28.43	19.57
OpenRSS-rank-2	26.61	21.78	35.77	25.98	32.60	23.51
OpenRSS-rank-4	25.38	21.63	36.53	27.73	29.64	20.49
OpenRSS-rank-8	27.26	22.51	37.07	26.28	35.23	26.06
OpenRSS-DLoRA(ours)	**30.73**	**22.81**	**37.88**	**28.58**	**37.25**	**27.55**

5 Conclusion

This work proposes the first open-vocabulary RGB-T semantic segmentation framework: OpenRSS. Our framework achieves the open-vocabulary capability by jointly utilizing the CLIP and the modified SAM model. In addition, when fusing RGB and thermal modality information, OpenRSS only needs to add a few learnable parameters to SAM to achieve effective multi-modal information fusion. Extensive experiments show that the proposed framework significantly outperforms previous methods on three RGB-T semantic segmentation benchmarks, validating the effectiveness and generalization of OpenRSS. Our work provides a feasible solution to open-vocabulary semantic segmentation with RGB-T images and a strong baseline for the community to advance future research.

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